

# Evaluation of an AI algorithm trained on an ethnically diverse dataset to screen a previously unseen population for diabetic retinopathy

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**Purpose:** This study aimed to determine the generalizability of an artificial intelligence (AI) algorithm trained on an ethnically diverse dataset to screen for referable diabetic retinopathy (RDR) in the Armenian population unseen during AI development. **Methods:** This study comprised 550 patients with diabetes mellitus visiting the polyclinics of Armenia over 10 months requiring diabetic retinopathy (DR) screening. The Medios AI-DR algorithm was developed using a robust, diverse, ethnically balanced dataset with no inherent bias and deployed offline on a smartphone-based fundus camera. The algorithm here analyzed the retinal images captured using the target device for the presence of RDR (i.e., moderate non-proliferative diabetic retinopathy (NPDR) and/or clinically significant diabetic macular edema (CSDME) or more severe disease) and sight-threatening DR (STDR, i.e., severe NPDR and/or CSDME or more severe disease). The results compared the AI output to a consensus or majority image grading of three expert graders according to the International Clinical Diabetic Retinopathy severity scale. **Results:** On 478 subjects included in the analysis, the algorithm achieved a high classification sensitivity of 95.30% (95% CI: 91.9%–98.7%) and a specificity of 83.89% (95% CI: 79.9%–87.9%) for the detection of RDR. The sensitivity for STDR detection was 100%. **Conclusion:** The study proved that Medios AI-DR algorithm yields good accuracy in screening for RDR in the Armenian population. In our literature search, this is the only smartphone-based, offline AI model validated in different populations.

**Key words:** Artificial Intelligence, deep learning, diabetic retinopathy, eye screening

Diabetic retinopathy (DR), a microvascular complication of diabetes mellitus (DM), is one of the leading causes of preventable blindness. It is estimated that 642 million people would be living with diabetes by 2040 worldwide.<sup>[1]</sup> The global prevalence of DR among people with diabetes is 34.6%, and it is 10.2% for sight-threatening diabetic retinopathy (STDR).<sup>[2]</sup> Over the past decade, the number of people with diabetes has increased.<sup>[1]</sup> Such high numbers not only pose a great economic burden but also create an ever-increasing demand for accessible eye care. Artificial intelligence (AI) can help bridge the otherwise widening gap between ophthalmologists and patients.

Current deep learning (DL)-based AI algorithms have shown performances approaching that of clinicians in detecting DR.<sup>[3–7]</sup> This encourages the deployment of such systems to reduce the burden on ophthalmologists. There is a requirement across all geographies to tackle the global problem of preventable blindness. It is thus important to focus on unbiased and robust

AI systems, which work equally well across ethnicities and populations.

Patient attributes, such as race/ethnicity, can introduce biases in AI systems, and these biases pose significant challenges in the development of AI-based models for DR screening. It is crucial for effective solutions deployed in different geographies to demonstrate consistent accuracy across diverse populations.<sup>[4,8,9]</sup> It should be noted that ethnic groups with darker skin tend to have higher melanin content within their uveal melanocytes, resulting in darker fundus pigmentation.<sup>[10,11]</sup> Consequently, although DR lesions are similar across all ethnic groups, the background color of the fundus can make these lesions more or less distinct. Thus, fundus pigmentation may also impact the interpretation of AI systems.<sup>[10]</sup>

We hypothesize that an AI algorithm can generalize across different populations. This should hold true even if the population is not represented in the training set. This assumption relies on an ethnically balanced and diverse

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dataset. The Medios AI-DR is an offline, smartphone-based algorithm trained on a diverse dataset that did not include Armenian eyes. We report here the accuracy of this algorithm on an Armenian population for the detection of referable diabetic retinopathy (RDR). The scope of the DR screening program in Armenia falls under the aegis of the Armenian Eye Care Project (AECF) team in cooperation with the Armenian Ministry of Health and the World Diabetes Foundation with an aim to end preventable blindness due to DR.<sup>[12]</sup> The AECF project is part of a larger initiative led by this team to ensure accessible eye care for all Armenians.

## Methods

This retrospective study was approved (IRB approval no.: N4-3/2020) by the institutional ethics committee. It was performed according to the tenets of the Declaration of Helsinki.

**Study Population Sampling and Imaging Protocol:** Patients with established diabetes attending the polyclinics of Armenia for DR screening were included in this study. The images were captured using the Fundus-on-Phone (FOP NM-10, Remidio Innovative Solutions), a smartphone-based fundus camera with a field of view of 40°, and collected over a 10-month period between July 2019 and April 2020. The study population included 550 consecutive subjects. Subjects with no established ground truth diagnosis were excluded.

**Retinal image acquisition:** The images were captured by non-medically trained operators in real-world conditions. The imaging protocol consisted of acquiring one disc and one macula-centered image per eye of each patient without dilation. This was performed during routine DR screening. Repeat images were captured when required to ensure sufficient image quality.

**Image Grading:** The consensus or majority image grading of three expert graders, two fellowship-trained ophthalmologists, and one certified optometrist formed the reference standard for assessing the AI algorithm. One grader, a fellowship-trained ophthalmologist, provided image-based diagnosis during the screening program. In total, 53 patients had no consensus after grading by the three graders. They were presented to two senior retina specialists, whose adjudicated diagnosis was deemed final. All graders were masked to the Medios AI-DR output and to each other's grading. The de-identified images were graded for the stage of DR and the presence of diabetic maculopathy. The International Clinical Diabetic Retinopathy classification was used to grade images. The images were graded as either no DR, mild non-proliferative DR (mild NPDR), moderate non-proliferate DR (moderate NPDR), severe non-proliferate DR (severe NPDR), proliferative DR (PDR), or ungradable. The graders also looked for hard exudates within 1 disc diameter of the fovea center. This was used as a surrogate marker for the presence of clinically significant diabetic macular edema (CSDME), which is considered a standard guideline in a screening context in the absence of stereo imaging.<sup>[13–15]</sup> An image was deemed ungradable if a reliable diagnosis of DR was not possible. This could happen in two distinct scenarios: 1) if major vessels could not be clearly identified, or 2) due to blurring, artifacts, under/over-exposure, or glare spanning half or more of the image. The patient-level diagnosis was inferred by the DR stage of the more affected eye. The consensus grading for each patient formed the final diagnosis. RDR was defined

as moderate NPDR or higher severity and/or the presence of CSDME. STDR was defined as severe NPDR or higher severity and/or the presence of CSDME.

**AI-based software architecture:** The Medios AI-DR consists of an ensemble of two convolutional neural networks based on the Inception-V3 architecture. It classifies color fundus images for the presence of RDR. The detailed software architecture has been previously published.<sup>[6]</sup> In brief, the training set consisted of 52,894 images as follows: 34,278 images were obtained from the Eye Picture Archive Communication System telemedicine program (EyePACS LLC, Santa Cruz, California). In total, 14,266 mydriatic images were captured using a Kowa VX-10α (Kowa American Corporation, CA, USA) at a tertiary diabetes center in India, and 4350 non-mydriatic images were taken in mass screening camps in India by using the Remidio FOPNM-10. The dataset was curated to contain as many referral cases as healthy ones.

**Automated image analysis:** The analysis was performed on the FOP NM10 with offline Medios AI-DR that does not require internet to provide a report. The de-identified images were loaded through Remidio's secure cloud connectivity system. The Medios AI-DR was manually run as per standard protocol. Following an automated analysis of image quality, each patient underwent an automated analysis for DR. The AI output No RDR or RDR, as well as the image quality analysis results, were noted. The "proceed anyway" override option was used for images that failed the AI quality check but received a consensus grading by the experts.

**Outcome measure:** The primary outcome measures were the sensitivity, specificity, false positives, false negatives, and predictive values (positive and negative) of the Medios AI-DR algorithm for detecting RDR on images captured using FOPNM-10 on this population. The secondary outcome measures were the sensitivity, specificity, and predictive values (positive and negative) of the algorithm for detecting any grade of DR. In addition, the sensitivity for detecting STDR was reported.

**Statistical analysis:** Considering a sensitivity of at least 80% with a precision of 10% and a prevalence of RDR of 20%, the required sample size was 308 patients for a 95% confidence interval.<sup>[16]</sup> We, however, looked at a larger sample size of 550 patients imaged over 10 months.

Next, 2 \* 2 confusion matrices were used to compute the sensitivity (true positive [TP] rate) and specificity (true negative [TN] rate) to detect RDR, any stage of DR, and STDR. The positive predictive value (PPV) and the negative predictive value (NPV) were evaluated. Furthermore, 95% confidence intervals (CI) were calculated for sensitivity, specificity, NPV, and PPV. The false positive (FP) rate was calculated as FP/FP + TN. The false negative rate was calculated as FN/FN + TP. The kappa statistic was used to determine the agreement between each expert grader to the consensus grading. A kappa value of above 0.8 was considered as high agreement, between 0.5 and 0.79 as moderate, and below 0.5 as poor agreement. Data were analyzed using the pandas (1.1.0), NumPy (1.19.5), and scikit-learn (0.23.1) libraries in Python (ver 3.7.7).

## Results

The mean age in this study cohort of 550 subjects was 61.6 ± 9.94 years (range: 12–83 years). Among them,

63.45% (n = 349) were females. In the final analysis, 478 subjects were included after excluding duplicate entries (n = 6) and patients deemed ungradable (n = 66) [Fig. 1].

Based on consensus image diagnosis by experts, any DR was present in 159 (33.26%) patients, RDR was seen in 149 patients (31.17%), and STDR in 62 patients (12.97%). The intergrader agreement (quadratic weighted kappa) between the individual certified experts and the majority diagnosis was moderately good (0.56–0.72).

In total, 478 subjects were fed to the AI to generate outputs for the presence or absence of RDR. The sensitivity for RDR was 95.30 % (95% CI: 91.9%–98.7%) and specificity was 83.89% (95% CI: 79.9%–87.90%). The NPV, that is, the probability of a subject with a negative screening test by the AI to truly not have RDR, was high (97.53%, 95% CI: 95.7%–99.3%). The key performance metrics for Medios AI-DR on the Armenian population are listed in Tables 1A and 1B.

The AI made a false diagnosis of RDR in 53 subjects. Four subjects had a consensus grading of mild NPDR, and the other 49 subjects had a consensus grading of no DR by the graders. Thus, the false positive rate was 16.11%. Among the 53 subjects, 17 were graded with AMD by at least one of the graders, while another nine were graded as having another pathology.

Table 1A: Confusion matrix to evaluate Medios AI- DR performance for RDR

| n=478           | Image grading<br>RDR positive | Image grading<br>RDR negative |
|-----------------|-------------------------------|-------------------------------|
| AI RDR positive | 142 (29.7%)                   | 53 (11.1%)                    |
| AI RDR negative | 7 (1.5%)                      | 276 (57.7%)                   |

All the seven subjects falsely diagnosed as RDR-negative cases by the AI were moderate NPDR cases. The false-negative rate was 4.70%. No cases of STDR were missed (100% sensitivity). Fig. 2 shows examples of subjects diagnosed correctly and incorrectly by the AI for RDR.

Discussion

This study showed the clinical effectiveness of the Medios AI-DR algorithm in detecting RDR in an Armenian population, reproducing its results from previous validation studies on the Indian population [Table 2]. The study highlighted the generalizability of this offline system in an ethnic population unseen during training. The study revealed excellent sensitivity (95.30%) and acceptable specificity (83.89%) for RDR.

The current study demonstrated a 100% sensitivity for sight-threatening DR. This subgroup entails patients at immediate risk of blindness if left untreated. Hence, this result supports the use of the Medios AI-DR as an aid in the early diagnosis of RDR. Thus, the specialists could focus on the treatment of patients with sight-threatening diseases rather than on screening. This is particularly valuable in regions with low density of ophthalmologists. This conclusion is further supported by a low false negative rate of 4.70% (7 cases of moderate NPDR missed), a parameter critical to denote the safety of the AI system. DR is understood to progress slowly in its early stages. The risk of missing such cases is mitigated by a recommendation for follow-up screenings every year.

The study showed 53 false positives (49 with no DR and 4 with mild NPDR). Furthermore, among the 53 subjects, 17 were graded with AMD by at least one of the graders, while another nine were graded as having another pathology. The accuracy is comparable to our findings on Indian cohorts.<sup>[6,7,17]</sup>

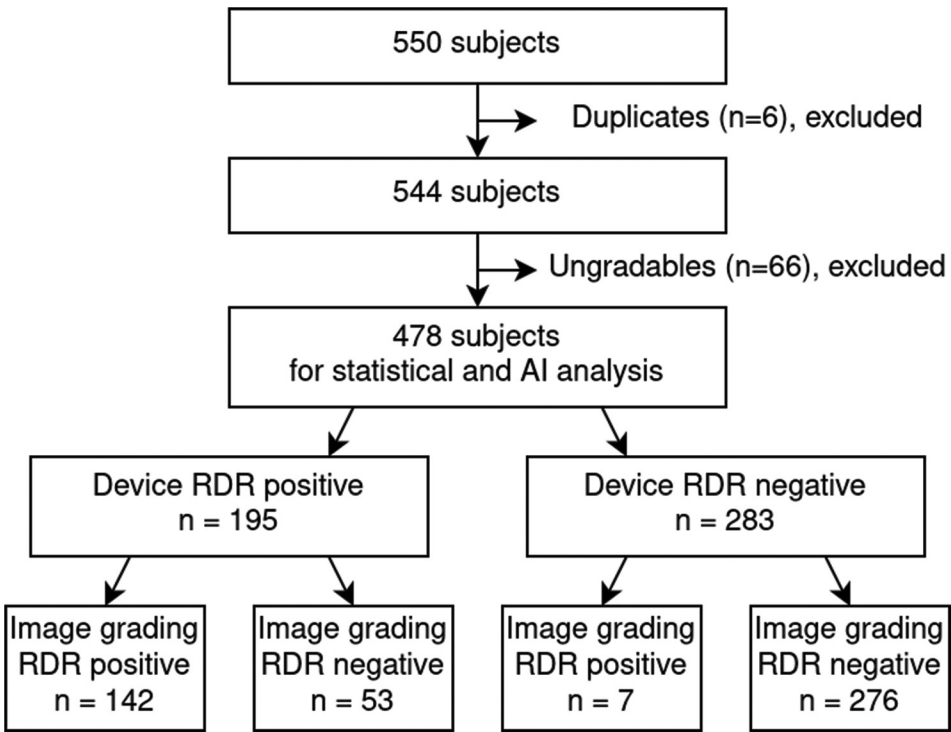
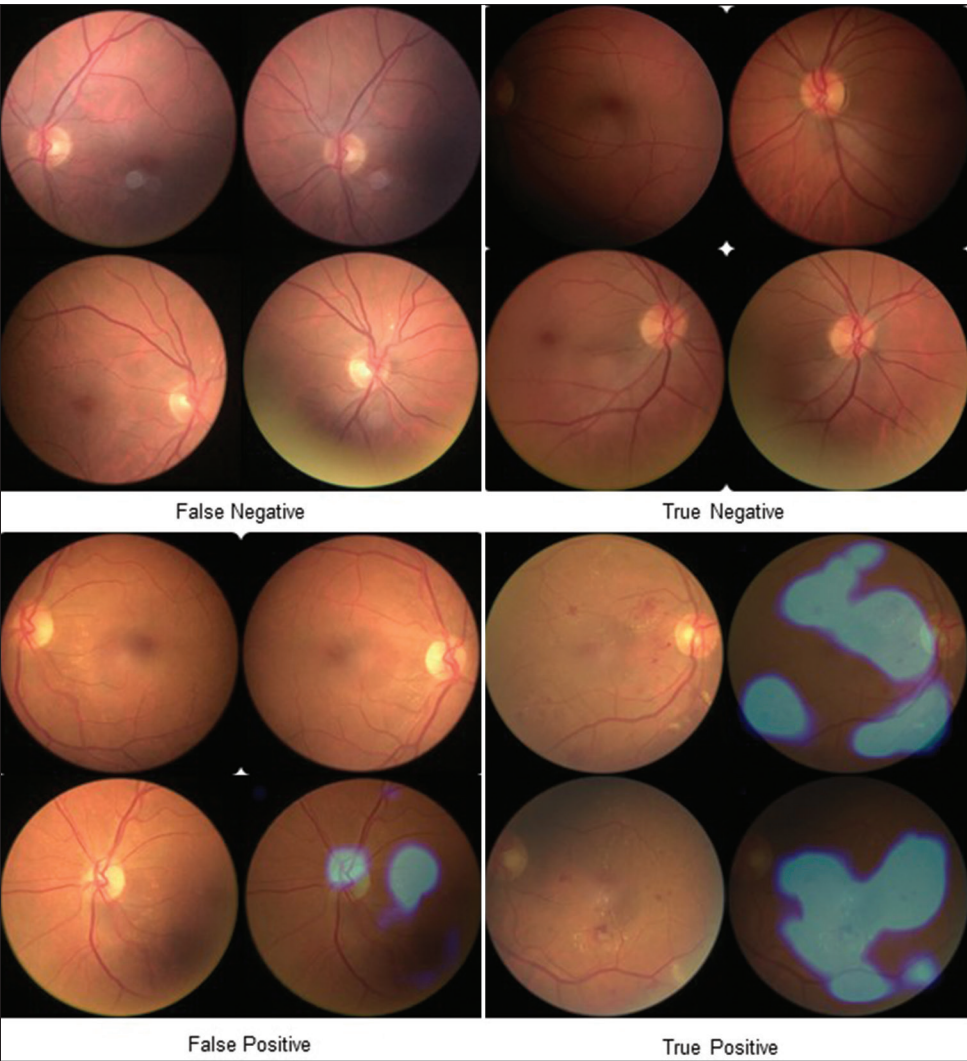


Figure 1: STARD Diagram for Medios AI-DR output for RDR





**Figure 2:** Examples of False Negative, True Negative, False Positive and True Positive patient images analyzed for RDR using Medios AI- DR. Activation maps are shown for images with positive AI diagnosis

| Table 1B: Performance of Medios AI-DR on the Armenian population |                      |                      |                  |
|------------------------------------------------------------------|----------------------|----------------------|------------------|
|                                                                  | RDR                  | Any DR               | STDR             |
| Sensitivity (95% CI)                                             | 95.30% (91.9%–98.7%) | 91.82% (87.6%–96.1%) | 100.00 (1.0–1.0) |
| Specificity (95% CI)                                             | 83.89 (79.9%–87.9%)  | 84.64% (80.7%–88.6%) | NA               |
| PPV (95% CI)                                                     | 72.82% (66.6%–79.1%) | 74.87% (68.8%–81.0%) | NA               |
| NPV (95% CI)                                                     | 97.53% (95.7%–99.3%) | 95.41% (93.0%–97.8%) | NA               |

| Table 2: Performance of Medios AI-DR in screening for RDR in previous studies |           |                              |                                |
|-------------------------------------------------------------------------------|-----------|------------------------------|--------------------------------|
| Study                                                                         | Ethnicity | Sensitivity                  | Specificity                    |
| Natarajan <i>et al.</i> <sup>[6]</sup>                                        | Indian    | 100.0% (95%CI: 78.2%–100.0%) | 88.4% (95% CI: 83.2%–92.5%)    |
| Sosale <i>et al.</i> <sup>[7]</sup> (SMART Study)                             | Indian    | 93% (95% CI: 91.3%–94.7%)    | 92.5% (95% CI: 90.8%–94.2%)    |
| Sosale <i>et al.</i> <sup>[17]</sup>                                          | Indian    | 98.84% (95% CI: 97.62%–100%) | 86.73% (95% CI: 82.87%–90.59%) |
| Current Study                                                                 | Armenian  | 95.30% (91.9%–98.7%)         | 83.89% (79.9%–87.9%)           |

We hypothesize that retinal pigmentation did not affect the system’s performance. The large and diverse dataset used during development ensured representation across dark to lightly pigmented retinas. Here, the misdiagnosis is rather

explainable by the presence of other pathologies being picked up by the AI (26/53, 49%) as referable diseases. We are currently in the process of deploying other disease-specific models such as AMD, which will address this in the near future. Graders,

however, only reported the status of DR. The presence of another retinal disease warranting a referral to a specialist and triggering a positive report by the AI thus cannot be ruled out. The false positive rate is within acceptable limits. The FDA-mandated superiority endpoints of 85% sensitivity and 82.5% specificity for an autonomous RDR AI were surpassed in this study.<sup>[14,15]</sup>

Previous studies have reported that a lack of ethnic diversity in training data impacted the performance of AI-based systems across ethnicities.<sup>[10]</sup> Varying concentrations of melanin in different ethnic groups affect the pigmentation of uveal melanocytes as well.<sup>[11]</sup> A different contrast between fundus and DR lesions due to a varying color of fundi may affect the performance of AI algorithms. AI systems should thus be trained on data from various ethnic groups to reduce bias toward a particular group. This is of particular importance when considering the deployment of systems across many geographies.

The Medios AI-DR had previously been validated in cohorts visiting primary and tertiary care centers in India only. Most studies relied on real-world images captured by minimally trained operators. They showed a sensitivity of 93%–100% and a specificity of 86.7%–92.5% [Table 2].<sup>[6,7,17]</sup> Though these studies demonstrated the consistency of the algorithm in different clinical settings and the community, these had not assessed its consistency across different ethnicities. The current study complements the earlier one and demonstrates generalizability to a new population, with good sensitivity and specificity. There are some reports of the use of ethnically diverse training datasets on the AI-based detection of RDR. Bellemo *et al.*<sup>[9]</sup> reported a sensitivity of 92.25% and a specificity of 89.04% in detecting RDR in an African cohort by using an AI algorithm trained with images predominantly from Chinese, Malay, and Indian populations. The system showed consistency and generalizability in detecting RDR in patients with dark fundi of the African population.<sup>[9]</sup> We are unsure if the good sensitivity and specificity were related to the AI training dataset that also included the Indian eyes. Both Indian and African groups rank higher in melanin synthesis and have nearly similar dark-colored fundi. Similarly, Ting *et al.*<sup>[4]</sup> reported a sensitivity of 90.5% and a specificity of 91.6% in identifying RDR in a multi-ethnic population. Again, these investigators had included images from Indian, Chinese, and Malay populations. The validation set further comprised African-American, Caucasian, and Hispanic populations with additional multi-ethnic validation datasets. All these studies indicate a common pattern: a sufficiently diverse training set can lead to generalizability beyond the ethnic groups included during training. While this hypothesis is reinforced in this study, we evaluated the AI performance in a different low-resource population that has significant accessibility issues for DR screening. To our knowledge, there is no peer-reviewed published evidence (MEDLINE literature search) on the performance of AI on images captured in a real-world diabetic screening program in Armenia. In addition, on the technical front, the Medios AI-DR is a lightweight ensemble architecture that utilizes a low processor environment of a smartphone-based fundus camera, allowing it to be deployed on the edge on the device. This could overcome barriers of internet connectivity and cloud-based inferencing in limited resource settings.

**Strengths:** First, the Medios AI-DR algorithm for RDR has been developed with over 50,000 images, which also included 34,278 images from the Kaggle-EyePACS dataset. The Kaggle-EyePACS dataset includes populations of indigenous American, African, European, Asian, and Indian subcontinent descent by design. It does not have any inherent issues with diversity or bias.<sup>[18,19]</sup> This could partly explain the absence of bias in this algorithm when used on both dark and lightly pigmented fundi. The results of this study further strengthened this statement. Second, this study truly captured real-world data where images were captured by minimally trained operators. Thus, the AI was subject to test on images of par quality unlike the pristine images obtained in the clinic where the performance will be far better.

**Limitations:** This is a retrospective study with inherent limitations attributable to any retrospective study. A notable limitation was the absence of a “live” AI quality check and feedback to the operator to recapture images. This implied that the study did not follow the exact protocol of two sufficient quality images per eye (one disc and one macula centered) in all patients, as required for optimum AI performance results. Instead, the study used all images captured (up to 7 images per patient) from multiple capturing attempts giving the worst-case scenario results. Inclusion of all available images might have also adversely affected the sensitivity and specificity. In addition, an accurate assessment of the image quality algorithm was not possible. A prospective trial would have been ideal to assess the best performance of the algorithm. The DR AI algorithm and image grading at least relied on two 40° fields of view per eye – one macula and one disc-centered image (covering approximately 60° field of view), potentially overlooking a few instances of severe DR extending beyond the captured area, particularly affecting the nasal region or extending beyond major blood vessel arcades. However, utilizing non-mydratic single or two-field fundus photography for DR screening aligns with the acceptable practices recommended by the International Council of Ophthalmology as well as the American Academy of Ophthalmology Guidelines.<sup>[20,21]</sup>

## Conclusion

There is evidence that a training set biased against a specific ethnic group does not generalize well beyond that group.<sup>[10]</sup> However, others and we have shown that an AI algorithm trained with an ethnically diverse dataset overcomes this deficiency.<sup>[4,9]</sup> In addition, we feel that additional investigations are needed to evaluate variations of fundus pigmentation across ethnicities and understand the similarities and differences to develop robust AI solutions by using retinal images.

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## References

- Ogurtsova K, da Rocha Fernandes JD, Huang Y, Linnenkamp U, Guariguata L, Cho NH, *et al.* IDF Diabetes Atlas: Global estimates for the prevalence of diabetes for 2015 and 2040. *Diabetes Res Clin Pract* 2017;128:40-50.
- Thomas RL, Halim S, Gurudas S, Sivaprasad S, Owens DR. IDF Diabetes Atlas: A review of studies utilising retinal photography on the global prevalence of diabetes related retinopathy between 2015 and 2018. *Diabetes Res Clin Pract* 2019;157:107840.
- Nielsen KB, Laurrup ML, Andersen JKH, Savarimuthu TR, Grauslund J. Deep learning-based algorithms in screening of diabetic retinopathy: A systematic review of diagnostic performance. *Ophthalmol Retina* 2019;3:294-304.
- Ting DSW, Cheung CY, Lim G, Tan GSW, Quang ND, Gan A, *et al.* Development and validation of a deep learning system for diabetic retinopathy and related eye diseases using retinal images from multiethnic populations with diabetes. *JAMA* 2017;318:2211-23.
- Rajalakshmi R, Subashini R, Anjana RM, Mohan V. Automated diabetic retinopathy detection in smartphone-based fundus photography using artificial intelligence. *Eye (Lond)* 2018;32:1138-44.
- Natarajan S, Jain A, Krishnan R, Rogye A, Sivaprasad S. Diagnostic accuracy of community-based diabetic retinopathy screening with an offline artificial intelligence system on a smartphone. *JAMA Ophthalmol* 2019;137:1182-8.
- Sosale B, Aravind SR, Murthy H, Narayana S, Sharma U, Gowda SGV, *et al.* Simple, mobile-based artificial intelligence algorithm in the detection of diabetic retinopathy (SMART) study. *BMJ Open Diabetes Res Care* 2020;8:e000892.
- Yip MYT, Lim G, Lim ZW, Nguyen QD, Chong CCY, Yu M, *et al.* Technical and imaging factors influencing performance of deep learning systems for diabetic retinopathy. *NPJ Digit Med* 2020;3:40.
- Belleemo V, Lim ZW, Lim G, Nguyen QD, Xie Y, Yip MYT, *et al.* Artificial intelligence using deep learning to screen for referable and vision-threatening diabetic retinopathy in Africa: A clinical validation study. *Lancet Digit Health* 2019;1:e35-44.
- Burlina P, Joshi N, Paul W, Pacheco KD, Bressler NM. Addressing artificial intelligence bias in retinal diagnostics. *Transl Vis Sci Technol* 2021;10:1-13.
- Hu DN, Savage HE, Roberts JE. Uveal melanocytes, ocular pigment epithelium, and Müller cells in culture: *In vitro* toxicology. *Int J Toxicol* 2002;21:465-72.
- Armenian EyeCare Project Diabetes Program - Armenian EyeCare Project. Available from: <https://eyecareproject.com/diabetes-program/>. [Last accessed on 2022 Mar 08].
- Diabetic eye screening: Professional guidance - GOV.UK. Available from: <https://www.gov.uk/government/collections/diabetic-eye-screening-commission-and-provide>. [Last accessed on 2022 Mar 08].
- Abràmoff MD, Lavin PT, Birch M, Shah N, Folk JC. Pivotal trial of an autonomous AI-based diagnostic system for detection of diabetic retinopathy in primary care offices. *NPJ Digit Med* 2018;1:39.
- Ipp E, Liljenquist D, Bode B, Shah VN, Silverstein S, Regillo CD, *et al.* Pivotal evaluation of an artificial intelligence system for autonomous detection of referable and vision-threatening diabetic retinopathy. *JAMA Netw Open* 2021;4:e2134254.
- Scanlon PH. The English National Screening Programme for diabetic retinopathy 2003-2016. *Acta Diabetol* 2017;54:515-25.
- Sosale B, Sosale AR, Murthy H, Sengupta S, Naveenam M. Medios- An offline, smartphone-based artificial intelligence algorithm for the diagnosis of diabetic retinopathy. *Indian J Ophthalmol* 2020;68:391-5.
- Cuadros J, Bresnick G. EyePACS: An adaptable telemedicine system for diabetic retinopathy screening. *J Diabetes Sci Technol* 2009;3:509-16.
- Huemer J, Wagner SK, Sim DA. The evolution of diabetic retinopathy screening programmes: A Chronology of retinal photography from 35 mm slides to artificial intelligence. *Clin Ophthalmol* 2020;14:2021-35.
- Williams GA, Scott IU, Haller JA, Maguire AM, Marcus D, McDonald HR. Single-field fundus photography for diabetic retinopathy screening: A report by the American Academy of Ophthalmology. *Ophthalmology* 2004;111:1055-62.
- Wong TY, Sun J, Kawasaki R, Ruamviboonsuk P, Gupta N, Lansingh VC, *et al.* Guidelines on Diabetic Eye Care: The International Council of ophthalmology recommendations for screening, follow-up, referral, and treatment based on resource settings. *Ophthalmology* 2018;125:1608-22.